

Review

Time-lagged panel models in psychotherapy process and mechanisms of change research: Methodological challenges and advances

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ABSTRACT

In recent years, there has been increasing interest in utilizing time-lagged panel models to study mechanisms of change in psychotherapy. These models offer valuable insights into the dynamic relationships between variables over time and offer stronger causal inference capabilities than cross-sectional analyses. Therefore, they are well-suited for modeling the intricate relationships between mechanisms of change and outcomes in psychotherapy studies, which are typically beyond experimental control. However, their complexity, coupled with the fact that detailed explanations are often embedded in dense statistical or econometric texts, poses challenges. This paper provides a background on cross-lagged panel models and delves deeper into explaining the issues of 1) dynamic panel bias, 2) long-run effects, and 3) testing whether different treatments work by different mechanisms. Using data from a psychotherapy study on treatment of adolescent depression, I demonstrate how these issues manifest in real data. In conclusion, I recommend using structural equation modeling to circumvent dynamic panel bias, reporting long-run effects to reveal the long-term impact of sustained therapeutic work on mechanisms of change, and carefully considering whether mediation, moderation, or a combination of both, best describes differential effects of mechanisms between treatments.

1. Introduction

In recent years, there has been an upsurge of interest in studying psychotherapy processes and mechanisms of change using time-lagged panel models (Flückiger et al., 2020; Rubel, Zilcha-Mano, Giesemann, Prinz, & Lutz, 2019; Zilcha-Mano, 2017; Zilcha-Mano & Webb, 2021). Conceptually, change mechanisms are intermediate changes in the client that lead to improvements in the ultimate outcome variable, while change processes are activities within sessions that facilitate these changes in mechanisms and ultimately the outcome (Doss, 2004). In this paper, I will primarily use the term ‘mechanisms’, but since my focus is on data analysis, the same arguments can be applied to processes. In the typical time-lagged panel study of mechanisms, one or more candidate mechanisms—such as cognitive distortions in cognitive therapy, or social support in interpersonal psychotherapy—are measured repeatedly alongside an outcome variable (e.g., symptoms, functioning, or quality of life) over the course of treatment. The researcher then uses a statistical model where earlier measures of the mechanism predict future values of the outcome variable, thereby accounting for the principle that causes must precede effects.

Reasons for this increased interest in time-lagged panel models are

probably manifold, but practical and ethical issues in experimentally manipulating processes and mechanisms generate a need for stringent alternative methods. As noted by Falkenström et al. (2017), time-lagged panel models are relatively strong for causal inference due to accounting for temporal precedence and some forms of unobserved confounding. Moreover, the results of these models can be communicated effectively to clinicians, since studying sequences in the form of ‘when the therapist does X, the patient reacts with Y’ fits well with how clinicians tend to evaluate their everyday work. Thus, statistical models that aggregate and statistically evaluate relationships among many such sequences resonate well with clinical thinking.

At the same time, time-lagged panel models are statistically complex, and these complexities are often misunderstood or overlooked by psychotherapy researchers. One reason for this, I contend, is that psychotherapy researchers tend to be trained in and used to think in terms of multilevel modeling (MLM; e.g., Snijders & Bosker, 2012), a framework highly relevant for certain longitudinal models (e.g., growth curve models) but less suitable for time-lagged panel models (Allison, 2015; Falkenström et al., 2022; Zyphur et al., 2019). In this paper, I further argue that structural equation modeling (SEM; e.g., Kline, 2016) is a more apt framework for time-lagged panel models. In the following, I

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will focus on a few issues regarding time-lagged panel models that are important for psychotherapy research. Some of these issues are buried in mathematically dense econometric texts or methodological papers requiring a lot of prior knowledge on statistics, while others are relatively new and have still not been generally integrated into psychotherapy research. Thus, while I do not claim originality for any of the issues discussed in this paper, there is a need for an introduction and explanation of time-lagged panel models focusing on application in psychotherapy research.

I begin by describing the basic cross-lagged panel model (CLPM), a term used for time-lagged panel models where two or more variables are repeatedly measured and prospectively predict each other, along with some variations of this model. In particular, I focus on the separation of within- and between-person sources of variance in CLPMs, which is an important recent development (Curran & Bauer, 2011; Hamaker, 2012; Hamaker et al., 2015). Following the work of Andersen (2022), I then introduce the distinction between observation-level and residual-level models in SEM, which represent two methods of separating within- from between-person variances. Given that MLM does not offer a similar distinction, it is likely that psychotherapy researchers may not have noticed or fully appreciated this concept.

The rest of the paper is then structured around three basic issues that I believe are often not sufficiently described in the context of psychotherapy research: 1) dynamic panel bias, 2) long-run effects, and 3) differential mechanisms between treatments. Each of these issues are described in separate sections. After this, I provide a psychotherapy research example to demonstrate each of these issues. The paper ends with some conclusions and reflections about time-lagged panel analysis in psychotherapy research. Throughout the paper I take the position that the primary aim of statistical models is to identify causal effects from data (see Zyphur et al., 2019 for a similar view). Therefore, I use causal language, which may feel unfamiliar to some readers (Grosz et al., 2020). However, it should be remembered that causal inference from these models is conditional on some assumptions, chief of which is the no unmeasured confounding assumption (Falkenström, Solomonov, & Rubel, 2023). Thus, although the aim is to identify causal effects, and time-lagged panel models are relatively strong for pursuing this aim, this does not imply that all causal inference challenges are resolved merely by employing these methods.

2. The Cross-Lagged Panel Model

Fig. 1 shows a path diagram for a simple CLPM. This path diagram is a graphical depiction of the following compact equations. For waves $t = 2$ up to and including the last wave ($t = T$):

$$Y_{i,t} = \mu_Y + \alpha_Y(Y_{i,t-1}) + \beta_{XY}(X_{i,t-1}) + \varepsilon_{Yi,t} \quad (1a)$$

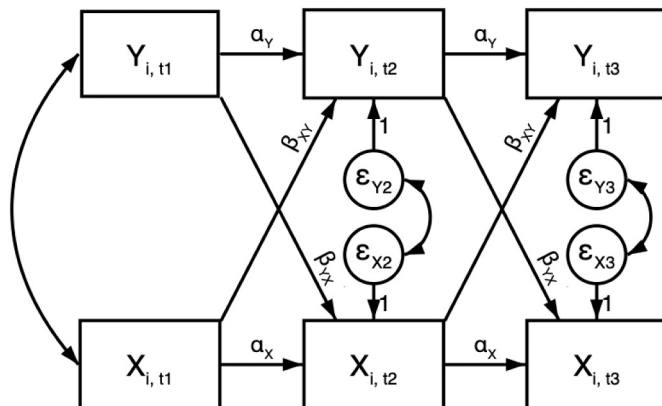


Fig. 1. Classic cross-lagged panel model.

$$X_{i,t} = \mu_X + \alpha_X(X_{i,t-1}) + \beta_{YX}(Y_{i,t-1}) + \varepsilon_{Xi,t} \quad (1b)$$

Eqs. (1a) and (1b) describe the relationships between continuous¹ variables $X_{i,t}$ and $Y_{i,t}$. In psychotherapy mechanisms of change research, $X_{i,t}$ may represent a mechanism of change such as a biological, cognitive, emotional, or behavioral variable presumed to be the active ingredient of treatment. Similarly, $Y_{i,t}$ might be an outcome variable, like symptoms, functioning, or quality of life. At wave one, $X_{i,1}$ and $Y_{i,1}$ function as independent predictor variables and are allowed to covary, as is typical in multiple regression models. This relationship is more evident in the path diagram, where no arrows point towards these two variables, but a double-headed arrow connects them. Additionally, error terms $\varepsilon_{Yi,t}$ and $\varepsilon_{Xi,t}$ are allowed to covary, as depicted by the curved arrows connecting them at each time point in the path diagram. At each subsequent wave, $Y_{i,t}$ and $X_{i,t}$ are regressed on their previous values (autoregression, indicated by α_Y and α_X), and the value of the other variable at the preceding time point (cross-lagged regression, captured by β_{XY} and β_{YX}). In this model, all regression coefficients are constrained to be equal over time (e.g., β_{XY} is the same between $X_{i,2}$ and $Y_{i,3}$ as between $X_{i,3}$ and $Y_{i,4}$, and so on, implying that the effect of the mechanism on the outcome variable is constant across treatment), although this is not a necessary requirement.

3. Separating between- from within-person variances in cross-lagged panel models

Panel data consists of repeated measures over time, for many individuals. This means that there are two dimensions in the data: the changes over time within individuals and differences between individuals. Several authors have argued that separating these dimensions in analyses is necessary, or the results will be difficult to interpret (Curran & Bauer, 2011; Hamaker, 2012; Wang & Maxwell, 2015). With long time-series (e.g., 50–100 repeated measures) it is possible to analyze each person's data separately, and then combine parameter estimates across individuals (Ramseyer et al., 2014). However, with short panels (e.g., fewer than 20 repeated measures, often only 3–5) correct estimation of parameters is unlikely, due to the limited number of data points (Lütkepohl, 2005). Isolating within-person fluctuations over time comes with the advantage that variables that are stable over the time of the study cannot, in principle, confound these within-person relationships (Moral-Benito et al., 2019; Zyphur, Allison, et al., 2019). Thus, stable characteristics of patients or therapists over the study period can be discounted as confounding variables, enhancing the potential for causal inference.

3.1. Two ways of separating between- from within-person variances in cross-lagged panel models

In psychotherapy research, the need to separate within- and between person variances when studying time-lagged panel models is by now reasonably well-established. However, it is likely that few researchers are aware that in SEM, there are two ways of separating between-person from within-person variances. These methods are sometimes referred to as *residual-level* and *observation-level* approaches (Andersen, 2022; Usami et al., 2019). In the residual-level approach, within- and between person variances are separated, and the cross-lagged panel model is estimated using the within-person residuals. For example, see Fig. 2a and Eq. 2 below, which illustrate the random intercept cross-lagged panel model (RI-CLPM; Hamaker et al., 2015):

$$Y_{i,t} = \mu_Y + Y_{Bi} + Y_{Wi,t} \quad (2a)$$

$$X_{i,t} = \mu_X + X_{Bi} + X_{Wi,t} \quad (2b)$$

¹ Extensions to categorical data exist, although these are beyond the scope of this article.

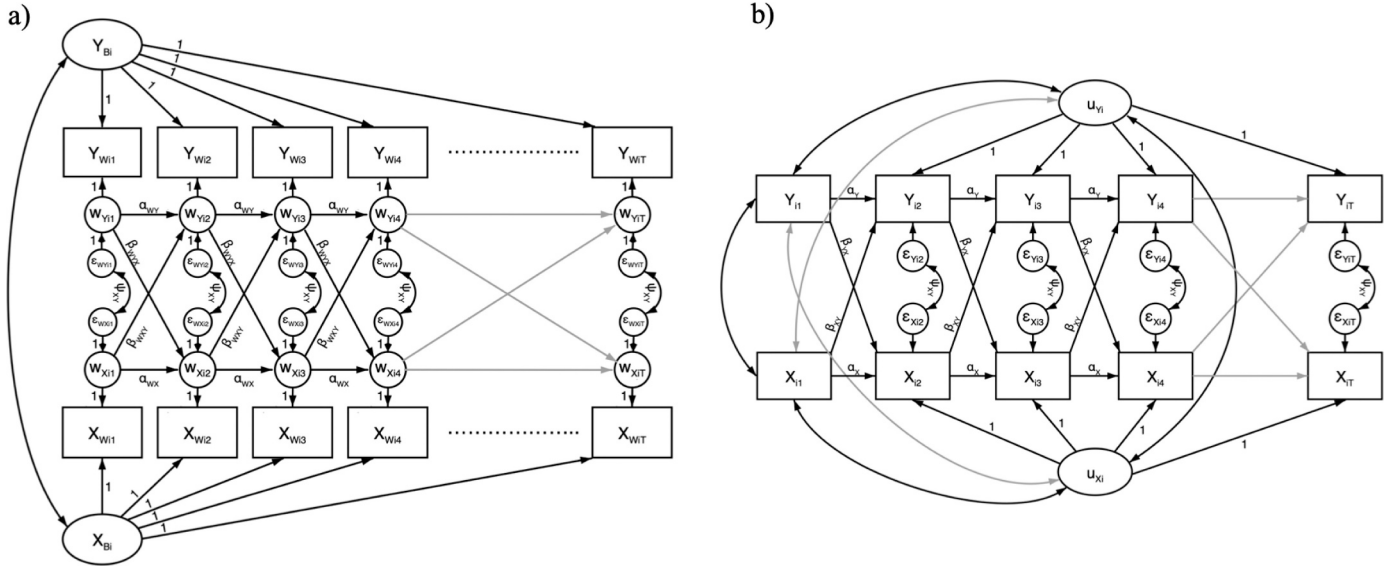


Fig. 2. Residual- and observation-level models for separating within- and between-person effects in cross-lagged panel models.

and

$$Y_{Wi,t} = \alpha_{WY}(Y_{Wi,t-1}) + \beta_{WXY}(X_{Wi,t-1}) + \varepsilon_{WYi,t} \quad (2c)$$

$$X_{Wi,t} = \alpha_{WX}(X_{Wi,t-1}) + \beta_{WTX}(Y_{Wi,t-1}) + \varepsilon_{WXi,t} \quad (2d)$$

In Eqs. (2a) and (2b), μ_Y and μ_X represent the overall means, while the between- and within-person variances of $Y_{i,t}$ and $X_{i,t}$ are separated into between- (Y_{Bi} and X_{Bi}) and within-person ($Y_{Wi,t}$ and $X_{Wi,t}$) variance components. In Eqs. (2c) and (2d), a cross-lagged panel model is specified for the within-person variance components. This variance-partitioning parameterization is similar to a MLM specification. However, it differs in being fully multivariate, as $X_{i,t}$ and $Y_{i,t}$ are modeled both as dependent variables and predictors simultaneously.

In contrast, in an observation-level model, between- and within-person variances are modeled directly on the observed variables. For waves 2 up to and including the last wave (T)²:

$$Y_{i,t} = \mu_Y + \alpha_Y(X_{i,t-1}) + \beta_{XY}(Y_{i,t-1}) + u_{Yi} + \varepsilon_{Yi,t} \quad (3a)$$

$$X_{i,t} = \mu_X + \alpha_X(X_{i,t-1}) + \beta_{YX}(Y_{i,t-1}) + u_{Xi} + \varepsilon_{Xi,t} \quad (3b)$$

The key difference between Eqs. (3a) and (3b) compared to Eqs. (1a) and (1b) is the addition of u_{Yi} and u_{Xi} . Like in the regular CLPM, at wave one, $X_{i,1}$ and $Y_{i,1}$ are treated as independent predictor variables that are adjusted for covariance with each other. To ensure that the within- and between-person effects of $X_{i,1}$ and $Y_{i,1}$ are modeled correctly, these variables must also be allowed to correlate with the between-person random terms u_{Yi} and u_{Xi} (see also Fig. 2b). Thus, at time $t = 1$, X and Y are treated as *predetermined*. This implies that we do not make any assumptions about their processes prior to the start of observation (Andersen, 2022).

3.1.1. Within-person effects

Andersen (2022) showed that when data were generated from a residual-level model, results from both residual- and observation-level models were identical on the within-person level (e.g., estimated auto- and cross-lagged regression coefficients, the α s and the β s in Eqs. 2 and

3). However, when data were generated from an observation-level model, within-person estimates from the residual-level model were biased. Andersen explained that in terms of residual-level models requiring stronger assumptions for *initial conditions*, that is, the assumptions for the studied processes before the measurements started. Specifically, the residual-level models require that the studied processes have reached *equilibrium*, so that means, variances, and covariances are all stable over time. In contrast, since in the observation-level models the between-person effects u_{Xi} and u_{Yi} are not separated from the auto- and cross-lagged effects, some forms of non-linearity in the initial changes in means and variances can be accommodated in these models. The reasons for this will be explained more below when we turn to between-person effects. For now, it is important to note that within-person estimates appear to be more robust in observation-level models.

3.1.2. Between-person effects

In Fig. 2, the between-person terms X_{Bi}/Y_{Bi} and u_{Xi}/u_{Yi} on the one hand, and the auto- and cross-lagged regressions (i.e., the α s and the β s) on the other, relate to each other in different ways in the observation- and residual-level models. Specifically, since in the residual-level model the auto- and cross-lagged regressions are modeled based on the residuals ($Y_{Wi,t}$ and $X_{Wi,t}$) after the between-person differences have been captured by X_{Bi} and Y_{Bi} , the within- and between-person processes are modeled separately. In the observation-level models, the auto- and cross-lagged effects are modeled directly among the observed variables, resulting in the between-person terms u_{Xi} and u_{Yi} impacting the observed variables not only directly but also indirectly via the auto- and cross-lagged paths. This can be seen most clearly in the path diagram depicted in Fig. 2b. For instance, starting at u_{Yi} and following the arrow pointing towards observed variable $Y_{i,2}$ (i.e., the factor loading for $Y_{i,2}$). From $Y_{i,2}$ there is one arrow pointing towards $Y_{i,3}$ (autoregression) and one pointing towards $X_{i,3}$ (cross-lagged regression). The factor loading arrow from u_{Yi} to $Y_{i,2}$ is in the same direction as the auto- and cross-lagged regressions pointing from $Y_{i,2}$ to $Y_{i,3}$ (i.e., $u_{Yi} \rightarrow Y_{i,2} \rightarrow Y_{i,3}$) and $X_{i,3}$ (i.e., $u_{Yi} \rightarrow Y_{i,2} \rightarrow X_{i,3}$). This constitutes a *chain* (Pearl, 2009), which implies that the presumed causal effect continues through $Y_{i,2}$ along the chain until there are no more arrows pointing towards other variables. This exemplifies one of the principles of *path tracing* (Wright, 1921), which also indicates that indirect effects can be calculated by multiplying the path coefficients along the chain.

In the residual-level model (Fig. 2a), if we start at Y_{Bi} and again follow the factor loading arrow towards observed variable $Y_{i,2}$, we see

² Predetermined observation-level models require one additional wave of data for model identification compared to residual-level models. For the models shown here, the residual-level model requires at least three repeated measures while the observation-level model requires at least four repeated measures.

that there is no arrow pointing away from $Y_{i,2}$. There is only an arrow pointing from $Y_{Wi,2}$ towards $Y_{i,2}$. This configuration, $Y_{Bi} \rightarrow Y_{i,2} \leftarrow Y_{Wi,2}$, forms a *collider*. Therefore, in this model, Y_{Bi} is assumed to only affect the observed variable at the same time point, with no indirect effects on later time points. If factor loadings are identical for all time points, as in the RI-CLPM, Y_{Bi} can be interpreted as the latent person-mean across all repeated observations.

In contrast, in observation-level models, the between-person estimates cannot be interpreted straightforwardly in this way. This is because, as seen earlier, between-person effects in these models have both direct and indirect effects (via the auto- and cross-lagged regressions) on observed variables. The latter has been termed *accumulated effects* (Usami et al., 2019), as the impact of between-person variables accumulates over time. Depending on the size and sign of the auto- and cross-lagged regressions, their effects later in the process may be larger than at the beginning due to the accumulation of many indirect effects. For instance, the effect of u_{Yi} on $Y_{i,3}$ in Fig. 2b is the sum of the direct effect $u_{Yi} \rightarrow Y_{i,3}$ and the indirect effect $u_{Yi} \rightarrow Y_{i,2} \rightarrow Y_{i,3}$. To get the effect of u_{Yi} on $Y_{i,4}$, there are three indirect effects in addition to the direct effect ($u_{Yi} \rightarrow Y_{i,4}$): $u_{Yi} \rightarrow Y_{i,2} \rightarrow Y_{i,3} \rightarrow Y_{i,4}$, $u_{Yi} \rightarrow Y_{i,3} \rightarrow Y_{i,4}$, and $u_{Yi} \rightarrow Y_{i,2} \rightarrow X_{i,3} \rightarrow Y_{i,4}$. In long panels, the number of indirect effects can be substantial. However, the longer the chains, the smaller the indirect effects tend to be, as they are calculated as the product of coefficients that are usually <1 (and greater than -1).

In one sense, this makes the between-person effects in the observation-level model harder to interpret, as noted by Jongerling and Hamaker (2011) and Usami (2021). However, this complexity may arise from researchers being more accustomed to the MLM variance-partitioning approach. If one adopts a causal inference perspective, observation-level models may even be more intuitive than the residualizing approach. From this perspective, the cross-lagged model is seen as a dynamic process that is affected by various kinds of *impulses* (Zyphur, Allison, et al., 2019; Zyphur, Voelkle, et al., 2019). The between-person effects u_{Xi} and u_{Yi} can be viewed as person-specific impulses, similar in shape but varying in size across individuals.

To take an example from psychotherapy research, consider a study on working alliance and psychological distress, where alliance and psychological distress are measured each session (e.g., Flückiger et al., 2020). In such a study, the between-person effect might be pre-existing trait-like characteristics (Zilcha-Mano, 2020) such as mentalization capacity, interpersonal skill deficits, or personality issues. The residualizing approach assumes these issues can be completely separated from the within-person cross-lagged process. However, observation-level models suggest that these person-level attributes impact observations on each occasion. This impact is then assumed to continue throughout the system of equations due to the auto- and cross-lagged regressions. From a causal inference perspective, it seems implausible that differences in mentalization, interpersonal skills, or personality problems would be entirely separate from within-person processes like the reciprocal influence between alliance and psychological distress. A more plausible scenario is that trait-like characteristics of the patient directly influence both alliance and distress on each measurement occasion, with this impact continuing indirectly via auto- and cross-lagged regressions.

Looking at Fig. 2a and b, we can think of Y_{Bi}/u_{Yi} and X_{Bi}/u_{Xi} as encompassing the influences of person-level or trait-like characteristics of patients, which affect the observations of alliance and psychological distress differently. In Fig. 2a, the influence of trait-like characteristics of alliance (X_{Bi}) on observations of alliance ($X_{i,t}$) is constant over time, but X_{Bi} does not influence psychological distress (although it is correlated with Y_{Bi} , the trait-level characteristic for psychological distress). In contrast, the model in Fig. 2b allows the trait-like characteristic of alliance to indirectly influence observations of psychological distress via the cross-lagged regressions (e.g., $u_{Xi} \rightarrow X_{i,t-1} \rightarrow Y_{i,t}$). In this specific example, allowing indirect effects seems more sensible than completely separating between- and within-person processes. However, the

appropriateness of such indirect effects varies depending on clinical theory, making the model's flexibility to allow for these effects valuable.

Between-person estimates from observation-level models may not be as straightforwardly interpretable as those from, for example, growth curve models. However, when viewed from the perspective of the whole model, it aligns more closely with clinical theory. Together with the increased robustness of within-person estimates demonstrated by Andersen (2022), I argue that the benefits of observation-level models significantly outweigh the complexity in interpreting between-person estimates. Next, the focus will shift to the issue of using MLM in estimating cross-lagged effects.

3.2. Why can we not use multilevel modeling to estimate cross-lagged panel models?

Psychotherapy researchers are often trained in MLM (e.g., Snijders & Bosker, 2012), a framework that is used for analyzing nested data, such as repeated measurements within persons in a panel data setup. MLM can be useful for growth curve modeling, for instance (Singer & Willett, 2003). However, researchers have cautioned against using MLM to estimate autoregression (Allison, 2015; Falkenström et al., 2017; Falkenström et al., 2022; Zyphur, Voelkle, et al., 2019). Autoregression is a critical component of a cross-lagged panel model, necessary for adjusting for pre-existing differences in variables at the previous time-point. This problem, known as “dynamic panel bias” or “Nickel's bias,” has been acknowledged since the early 1980s (Nickell, 1981). While mathematical derivations have been used to explain this bias, more intuitive explanations are hard to find in the literature. In the following sections, I will first explain dynamic panel bias using the traditional account, then attempt a more intuitive explanation.

3.3. Dynamic panel bias

A dynamic panel model is a model in which the effect of a lagged version of the dependent variable, Y , is included as a predictor to adjust for pre-existing differences at the previous time-point (e.g., Rabe-Hesketh & Skrondal, 2012). If a lagged dependent variable is included in a model that also adjusts for between-person differences, a typical multilevel model equation might be represented as follows:

$$Y_{i,t} = \mu + \beta_X(X_{i,t-1}) + \alpha_Y(Y_{i,t-1}) + u_i + \varepsilon_{i,t} \quad (4)$$

Although Eq. (4) separates within- from between-person variances for $Y_{i,t}$ using the random intercept u_i , it assumes that predictors $X_{i,t-1}$ and $Y_{i,t-1}$ contain no between-person variance (Firebaugh et al., 2013). This assumption is highly unrealistic, especially for $Y_{i,t-1}$ given that we have included u_i precisely because we want to adjust for between-person variance in $Y_{i,t}$. Obviously, if $Y_{i,t}$ consists of both within- and between-person variances, $Y_{i,t-1}$ will also do so. This leads to endogeneity, a violation of the fundamental regression assumption that predictors must be independent of error terms (Wooldridge, 2010). In this case, the predictor $Y_{i,t-1}$ is not independent of u_i , the random intercept, which acts as a Level-2 error term. To address this, most multilevel modelers apply person-mean centering, which involves subtracting each person's mean across all time-points from each observation, leaving only fluctuations around a mean of zero (Raudenbush & Bryk, 2002; Wang & Maxwell, 2015):

$$Y_{i,t} = \mu + \beta_{XY}(X_{i,t-1} - \bar{X}_i) + \alpha_Y(Y_{i,t-1} - \bar{Y}_i) + u_i + \varepsilon_{i,t} \quad (5)$$

In this equation, $X_{i,t-1}$ and $Y_{i,t-1}$ are person-mean centered and do therefore not contain any between-person variation. Therefore, they are independent of the random intercept u_i , seemingly resolving the issue. However, the challenge for intuitive understanding is that person-mean centering $Y_{i,t-1}$ merely shifts the dependence from $Y_{i,t-1} \leftarrow u_i$ to $Y_{i,t-1} \leftarrow \varepsilon_{i,t}$. This occurs because $\bar{Y}_i$ is part of both $Y_{i,t-1}$ and $\varepsilon_{i,t}$, inducing a negative correlation between $(Y_{i,t-1} - \bar{Y}_i)$ and $\varepsilon_{i,t}$ when $\bar{Y}_i$ is subtracted

from $Y_{i,t-1}$. Consequently, the model still suffers from endogeneity, and the coefficient α_Y for $Y_{i,t-1} - \bar{Y}_i$ will be biased, as extensively documented in the literature (Zyphur, Voelkle, et al., 2019). The bias to α_Y can be calculated using the following formula:

$$-\left(\frac{1 + \alpha_{YP}}{T - 1}\right) \quad (6)$$

In Eq. (6), α_{YP} is the population coefficient for lagged Y , and T is the number of repeated measures. Thus, the bias for α_Y , the coefficient of lagged Y , is always negative. This means that true positive autoregression is underestimated (becomes smaller), while zero or negative autoregression is overestimated (shifts towards more negative values). Additionally, the bias decreases with longer time series (since T is in the denominator) and increases as the population autoregression, α_{YP} , becomes larger (as it is in the numerator).

Although this bias is concerning in itself, some researchers may think it is not a significant problem, since the effect of lagged Y is typically used as a covariate only. However, this perspective overlooks the fact that all coefficients in a model are estimated simultaneously and are interdependent. Therefore, if α_Y in Eq. (5) is biased, it is likely that β_{XY} , the coefficient of primary interest, is also biased. Indeed, Falkenström et al. (2022) demonstrated in a simulation study that, under certain conditions, the bias in the coefficient for the lagged focal predictor (β_{XY}) was so substantial that the estimated coefficient was statistically significant and of a similar magnitude in the opposite direction from the true effect.

3.4. A more intuitive explanation of dynamic panel bias

Although the previous section may have convinced some readers of the existence of dynamic panel bias, many will probably find the explanation hard to grasp intuitively. In the following, I will present an alternative explanation that may be more accessible to readers less versed in mathematics. The issue with person-mean centering arises from the assumption that a person's observed mean score is equal to their true mean, without any measurement error (Lüdtke et al., 2008). However, according to classic test theory, this is highly unlikely, particularly with a small number of observations, as is the case in panels with few repeated measures. In practice, when subtracting the observed person-mean from each of that person's scores, we are subtracting not only the true mean but also random measurement error. To demonstrate the effect of this, let's consider an example³: We generate three random variables, $Y_{i,1}$, $Y_{i,2}$, and $Y_{i,3}$, representing individual i at times 1, 2, and 3, respectively. These variables are independent from each other ($r = 0.00$), with a mean of zero and the same standard deviation across the three occasions. Now, we subtract the person-mean (i.e., the row mean) across the three occasions from these variables. Although the true person mean is zero, with a large enough sample size, the correlations among the transformed variables $Y_{i,1} - \bar{Y}_i$, $Y_{i,2} - \bar{Y}_i$, and $Y_{i,3} - \bar{Y}_i$ will be $r = -0.50$! Interestingly, this observed bias aligns perfectly with the formula for dynamic panel bias, as detailed in Eq. (6): $-\left(\frac{1+\alpha}{T-1}\right) = -\left(\frac{1+0}{3-1}\right) = -\left(\frac{1}{2}\right) = -0.50$.

To understand this phenomenon, it's important to consider that the means we're subtracting are comprised of random noise that happen to be consistently present across all three variables. Effectively, subtracting this consistent random noise across the variables results in the induction of a negative correlation. The reason for this is easiest to see if we have only two repeated measures. The person-mean centered version of these two measures will correlate -1.0 , since if one is positive the other one will always be negative. Thus, we have established that person-mean centering introduces negative autocorrelation into the centered variable. The next step involves inserting a lagged version of this variable as

a predictor in a regression model for the original variable, $Y_{i,t}$. Whether the model is a standard linear regression or a random intercept two-level model, the outcome remains consistent with a correlation of $r = -0.50$, assuming the sample size is sufficiently large.

As previously mentioned, the primary research interest often lies in the cross-lagged effect, β_{XY} in Eq. (5). The bias affecting this coefficient is more complex to derive than the bias in α_Y . Unlike α_Y , which is biased even when all covariances are zero, β_{XY} is estimated accurately under these conditions. However, such a scenario with zero covariances is probably rare in real data. Bias in β_{XY} arises from a complex interaction among the covariances of the included variables. In principle, it can be determined through path tracing, though the calculations grow increasingly complex with each added wave of data. Falkenström et al. (2022) demonstrated that bias for β_{XY} was most influenced by reverse cross-lagged effects in the data, that is, $Y_{i,t-1} \rightarrow X_{i,t}$ as opposed to $X_{i,t-1} \rightarrow Y_{i,t}$. For instance, when data was generated with $T = 5$ and a reverse cross-lagged effect of 0.20, without other auto- or cross-lagged effects, estimates from Eq. (5) showed $\alpha_Y = -0.25$ and $\beta_{XY} = -0.05$, both of which should have been zero. Though $\beta_{XY} = -0.05$ might seem minor, it led to a Type-I error rate of 21%, significantly higher than the acceptable nominal rate of 5%. Under certain conditions, the bias in β_{XY} was as large as -0.34 , implying a risk of incorrectly concluding that X causes Y when in reality Y is causing X .

3.5. Solutions to dynamic panel bias

There are three known solutions to dynamic panel bias: 1) latent variable models (Moral-Benito et al., 2019), 2) instrumental variable regression (Arellano & Bond, 1991), and 3) bias correction procedures (Breitung et al., 2022; Bun & Carree, 2005). Instrumental variable regression is complex and requires precise specification; otherwise, the results may be as biased, or more so, than without any adjustment (Kiviet, Pleus, & Poldermans, 2015). Bias correction has been used primarily in econometrics, using models that are prevalent there. In psychology, latent variable SEM is the preferred method, and I recommend it due to its multiple advantages and relatively few disadvantages, especially compared to the instrumental variable method (Moral-Benito et al., 2019). Using latent variables to separate within- and between-person variances avoids dynamic panel bias, as it achieves separation in a single step without assuming that observed means are error-free (Lüdtke et al., 2008).

An exception to the general rule that time-lagged dynamic panel models should be estimated using SEM is the Dynamic Structural Equation Modeling (DSEM; Asparouhov et al., 2018) framework. DSEM is a hybrid modeling framework that combines features of MLM (using data in long format) and SEM. DSEM employs the latent covariate approach (Lüdtke et al., 2008) to separate within- and between-person variances for the predictor, meaning that random effects are added to both the dependent variable Y (as in regular MLM) and the predictor X . Additionally, DSEM estimates autoregression using latent variables, effectively circumventing dynamic panel bias. However, DSEM is grounded in time-series analysis and thus assumes a large number of repeated measures (Schultzberg & Muthén, 2018), with $T = 10$ as the absolute minimum. For data with shorter T , regular SEM with variables in wide format is more appropriate.

4. Long-run effects

A common misunderstanding about cross-lagged effects is that they are limited to short-term impacts, such as session-to-session effects of a process or mechanism on symptoms. The argument is that while such effects are informative for establishing causal inference through temporal precedence, they do not reveal anything about the ultimate outcome of therapy. For instance, in studying the impact of working alliance on changes in psychological distress, session-to-session variations in the working alliance quality may predict subsequent

³ The R code for this demonstration is included in the Online Supplement.

improvements or deteriorations in distress (Flückiger et al., 2020). However, this does not necessarily indicate who will fare better at the end of treatment. It is even possible that patients with the most significant session-to-session effects are those with the poorest overall outcomes (Crits-Christoph & Connolly Gibbons, 2021). An example is the session-to-session effects of emotion regulation on symptoms. This effect might be most pronounced in patients with borderline personality disorder, who often experience significant fluctuations in both emotion regulation and symptoms, yet may have worse overall treatment outcomes.

Although the previously mentioned points are likely true, it is incorrect to conclude that time-lagged effects are of little importance for final outcome. Such a conclusion is likely reached by researchers accustomed to classic mediation analysis, which aims to explain between-group outcome differences between experimental and control groups.⁴ In contrast, a cross-lagged panel model is based on the (co) variances over time in predictor and outcome variables. This means the model's results indicate the effects of a *hypothetical* change in one variable on the other variable at later points in time. For instance, while borderline patients in the given example might 'drive' the results of the analysis of emotion regulation on symptoms, these results can still inform us about emotion regulation as a potential mechanism of change, particularly for those patients driving the analysis. Stated differently, in this example, borderline patients do not *confound* the analysis of emotion regulation and symptoms. Instead, they *moderate* it, showing stronger within-person time-lagged effects compared to other patients.

Moreover, the hypothetical effect of variable X on variable Y extends beyond the immediate cross-lagged effect from one time-point to the next. As illustrated in Fig. 1, although we only observe lag-1 effects of $X_{i,t-1}$ on $Y_{i,t}$, indirect effects of X on Y at subsequent time-points (i.e., $t + 1$, $t + 2$ etc.) also occur due to autoregressions and reverse cross-lagged effects. Therefore, to understand the total effect of X on Y over time, both direct and indirect effects can be summarized to determine the overall impact. This is the basis for the long-run effect, or long-run multiplier. Shamsollahi et al. (2021) describe several types of long-run effects, but here I focus on the classic type, which is based on a *hypothetical one-unit increase in X that remains stable over time*. For instance, in the emotion regulation and symptoms example, the long-run effect would represent the anticipated improvement in symptoms if emotion regulation improved by one unit and maintained at that level over an extended period. Although this might seem abstract, it is what we regularly do in regression analysis when we interpret the regression coefficient as the expected change in Y from a one-unit increase in X .

To illustrate the long-run effect, Fig. 3 presents a simplified⁵ cross-lagged model with only paths from $X_{i,t-1}$ to $Y_{i,t}$, and no autoregression in X (autoregression is present only in Y). In this model, the cross-lagged effect from $X_{i,t-1}$ to $Y_{i,t}$ is set to 0.10, and the autoregressive effect in Y to 0.50. These parameter values are arbitrary but are reasonably estimates based on my experience from psychotherapy research studies. If we take $Y_{i,3}$ as the starting point, this variable is influenced by both $Y_{i,2}$ and $X_{i,2}$. The direct effect of $X_{i,2}$ on $Y_{i,3}$ is 0.10. Additionally, there is an indirect effect of $X_{i,1}$ on $Y_{i,3}$ via $Y_{i,2}$, calculated as $0.10 (X_{i,1} \rightarrow Y_{i,2}) \times 0.50 (Y_{i,2} \rightarrow Y_{i,3})$, which equals 0.05. Thus, the total effect of $X_{i,t}$ on $Y_{i,3}$ is 0.15 ($0.10 + 0.05$). For longer panels, this summing of direct and indirect effects continues: $0.10 \times 0.50^2 = 0.025$, then $0.10 \times 0.50^3 = 0.0125$, and so on. With four waves, the total effect is 0.175 ($0.15 + 0.025$), and with five waves, it is 0.1875 ($0.175 + 0.0125$).

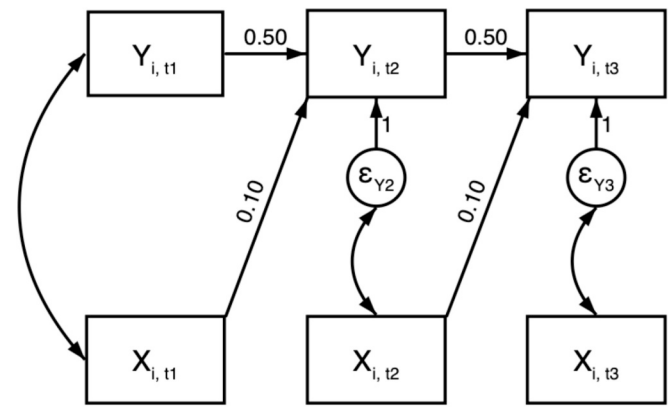


Fig. 3. Unidirectional dynamic panel model for illustrating the calculation of long-run effect.

Fig. 4 shows a graph of the results from summing effects for up to ten waves of data. As depicted in the graph, the increase in the total effect tends to flatten out around 8–10 waves, with the total effect approaching 0.20. This value, in this case 0.20, represents the long-run effect. In a model where both cross-lagged and autoregression effects are time-invariant, the long-run effect can be calculated as $\frac{\beta}{1-\alpha}$ (see Shamsollahi et al., 2021, p. 441) with β representing the cross-lagged effect and α the autoregression effect (for Y). For this example, it calculates to $\frac{0.10}{1-0.50} = 0.20$, which matches the value obtained by summing the direct and indirect effects over time. As the formula shows, a larger autoregression coefficient results in a larger long-run effect, since it reduces the denominator. This happens because a larger autoregression effect prolongs the persistence of the cross-lagged effect, delaying the indirect effects' convergence to zero.

The long-run effect implies that cross-lagged effects can extend beyond short-term predictions from one time-point to the next, potentially accumulating over time to produce a larger effect at longer intervals, depending on the magnitude of autoregression in Y . Incidentally, this helps explaining why cross-lagged effects are often much smaller than cross-sectional regression coefficients or correlations. For instance, Orth et al. (2022) found that the average standardized cross-lagged effect was 0.07 for the classic CLPM, and 0.05 for the RI-CLPM. Based on these findings, the authors suggest benchmarks for cross-lagged effects of 0.03 (small), 0.07 (medium), and 0.12 (large), which differ significantly from the Cohen (1988) benchmarks for correlations (small = 0.10, medium = 0.30, large = 0.50) and the Gignac and Szodorai (2016) empirically derived benchmarks (small = 0.10, medium = 0.20, large = 0.30). However, comparing a short-term cross-lagged effect (perhaps from one week of therapy to the next), adjusted for the previous outcome variable value, with cross-sectional

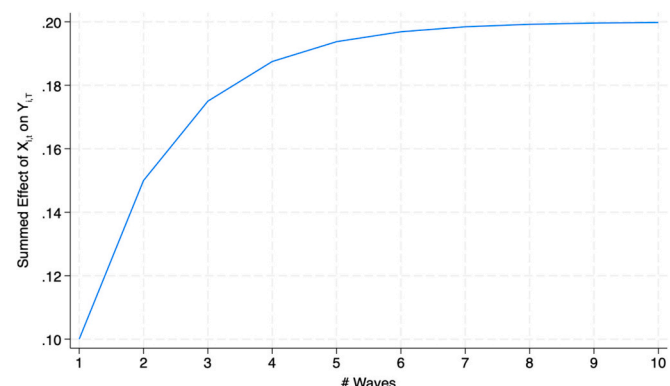


Fig. 4. The sum of direct and indirect effects approaching the long-run effect.

⁴ Modern mediation analysis, as described by Hayes (2009), is not focused on explaining mean differences between experimental and control groups, but rather on estimating the indirect effects of treatment on outcomes through mediators. This also means that a mean difference in outcomes between groups is not a prerequisite. The indirect effect is calculated based on covariances rather than means, aligning with the argument I present in this paper for cross-lagged panel models.

⁵ For pedagogical reasons, the model is depicted without separation of within- from between-person variances, but the principle for calculating the long-run effect is the same.

correlations may not be a meaningful comparison.

The formula for the long-run effect demonstrates that even a small cross-lagged effect can accumulate into a substantial impact over a longer period, particularly when autoregression is strong. Furthermore, the long-run effect suggests that cross-lagged effects might not merely reflect short-term fluctuations during treatment but could indicate a mechanism of change that, if consistently improved, may lead to lasting and meaningful psychotherapeutic outcomes. This also implies that the effect sizes of cross-lagged coefficients should not be assessed without considering autoregression. Given that the long-run effect is a synthesis of cross-lagged and autoregressive effects, it is advisable for researchers to routinely report long-run effects in psychotherapy studies.

4.1. Testing whether different treatments work by different mechanisms

In psychotherapy research, there is considerable interest in determining whether different treatments, such as treatment A and treatment B, operate through distinct mechanisms. For example, in a study comparing cognitive-behavioral therapy (CBT) to psychodynamic therapy (PDT), it might be posited that, even despite similar outcomes, CBT and PDT work through different mechanisms. The method generally recommended for analyzing mechanisms of change is mediation analysis (Kazdin, 2007), with the mediator variable operationalizing the theorized mechanism of change. However, when employing within-person time-lagged panel models, researchers often resort to moderation tests to explore hypotheses concerning differential mechanisms between treatments. This practice likely stems from the difficulty conceptualizing a model in which the predictor is a between-person variable (the treatment variable), while the mediator is a within-person variable that varies only within individuals, given that within-person effects are, by definition, unaffected by between-person differences. Consequently, moderation may appear to be the only possible approach for examining between-person differences in within-person effects (Falkenström et al., 2017).

A common approach to studying differential mechanisms is adding an interaction term between treatment type and a candidate mechanism to see if the mechanism's effect varies between treatments (Falkenström et al., 2013; Flückiger et al., 2020). For instance, a researcher may hypothesize that cognitive change is a specific mechanism in CBT. To test this, they might predict a stronger cross-lagged effect of cognitions on depressive symptoms in CBT compared to PDT, since PDT theory does not specifically focus on dysfunctional cognitions. On the face of it, this seems logical since a significant interaction between dysfunctional cognitions and treatment would mean a difference in the effect of dysfunctional cognitions between CBT and PDT on depressive symptoms. However, this approach implicitly assumes that the *causal relationship* between cognitions and depressive symptoms differs based on whether a patient is in CBT or PDT. In other words, it suggests that in CBT, a shift towards more realistic or positive cognitions about the self, world, and future leads to symptom improvement. However, in PDT, a similar cognitive improvement supposedly does *not* affect depressive symptoms. This implies that any natural or indirect cognitive changes in PDT patients, perhaps due to a focus on defense restructuring or transference interpretation, would not contribute to symptom improvement.

Framed this way, the moderator hypothesis appears less convincing. The fact that a PDT therapist does not actively focus on changing cognitions does not mean that such changes, if they occur, would not benefit the patient in reducing depressive symptoms. When considering differential mechanisms of change in psychotherapy, I believe we often have a simpler idea in mind. Specifically, the idea is that various treatments influence different naturally occurring cause-effect relationships, with the distinction between orientations lying in which mechanisms are deemed most potent and thus emphasized. However, since most psychotherapy researchers are trained in MLM, where testing for moderation is straightforward, this has become the predominant method in our field.

Before introducing the alternative model I propose, it is important to acknowledge a potential caveat. In some cases, it may indeed be plausible that the treatment context alters the causal relationships between variables. For example, the role of the working alliance may vary fundamentally between CBT and PDT. In CBT, the alliance is often seen as a prerequisite for the effectiveness of CBT techniques. Conversely, in PDT, especially with certain patients, the alliance is viewed as a change process in its own right (Zilcha-Mano et al., 2016). However, in many cases (such as the cognition → depression link), it seems highly implausible that the causal structure between variables would shift merely due to the use of a different treatment approach.

4.2. A mediation model of differential mechanisms of change

An alternative to moderation is to hypothesize that the causal relationship between a mechanism (e.g., cognitions/emotion regulation) and an outcome (e.g., depression) is inherent in nature. The distinction between treatments lies not in changing the causal relationships between variables, but in whether a treatment *utilizes* a specific mechanism. Thus, a treatment emphasizing cognitions as a mechanism of change is expected to have a stronger impact on cognitions than a treatment not focused on this mechanism. This implies a mediation model of the treatment's effect on the outcome via the mechanism, rather than a moderation model. Fig. 5 illustrates such a mediation model. In this model, the cross-lagged paths from M (Mechanism) to Y (Outcome) are not expected to differ based on Treatment, unlike in a moderator model. Instead, Treatment is hypothesized to influence $M_{i,t}$ (the candidate mechanism), which then affects $Y_{i,t}$ (outcome). The indirect effect of Treatment on Outcome via the Mediator can be calculated by multiplying the effect of Treatment on the Mediator (the 'a' path in Fig. 5) with the effect of the Mediator on Outcome (the 'b' path). Bootstrapped confidence intervals are used for testing statistical significance, as is common in mediation models (Hayes, 2009). If path coefficients are time-invariant, a single test suffices; otherwise, separate tests are needed for each time-point.

Thus, even though within-person effects cannot be confounded with between-person differences, it is still possible for between-person influences to be mediated by within-person effects. In Fig. 5, Treatment is specified as affecting the mediator M (path 'a'), which in turn affects the outcome Y at the following time point via the within-person cross-lagged effect (path 'b'). The mediation model is grounded in assumptions about mechanisms of change that differ from those in the moderation model. In this model, the path from Mediator to Outcome is assumed to be consistent across treatments. However, the influence of Treatment on the Mediator is stronger in treatments focused on the mechanism represented by the Mediator. Thus, it is not the causal effect of the mechanism on outcome that differs between treatments, but how much treatment is focused on changing the mechanism.

This model can be integrated with the concept of long-run effects, as discussed in the previous section. Specifically, we may want to determine the proportion of the total outcome attributable to the indirect effect of Treatment via the Mediator (Goldsmith et al., 2018). To do this, we calculate the total indirect effect by summing all indirect paths from Treatment to the final outcome $Y_{i,T}$ that originate from any of the 'a' paths (Treatment → $M_{i,1}$ and Treatment → $M_{i,2}$ in Fig. 5). In the three-wave example shown in Fig. 5, the number of indirect effects is limited, including Treatment → $M_{i,1}$ → $Y_{i,2}$ → $Y_{i,3}$, Treatment → $M_{i,2}$ → $Y_{i,3}$, and Treatment → $M_{i,1}$ → $M_{i,2}$ → $Y_{i,3}$.⁶

⁶ As the number of waves increase, the number of indirect effects grow exponentially, and it quickly becomes hard to keep track of all of them. Therefore, if a researcher wants to calculate this effect, some kind of automatization is needed. Fortunately, there is an R package ('manymome') that helps with this.

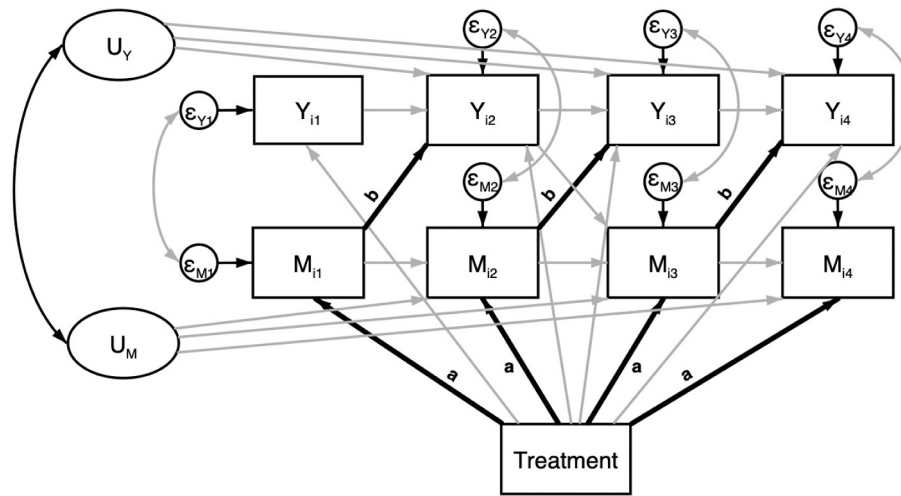


Fig. 5. Hypothesized mediation model of treatment on outcome via mechanism. Note. Mediation is calculated as $a \times b$.

5. Psychotherapy research example

To illustrate these issues, I utilized data from a randomized trial comparing two forms of internet-based psychotherapy for depressed adolescents: cognitive-behavioral therapy and psychodynamic therapy (Mechler et al., 2022). The study was a non-inferiority trial, comparing the newly developed internet-based psychodynamic treatment (IPDT; Lindqvist et al., 2020) to the more established internet-based cognitive-behavioral therapy (ICBT; Topoco et al., 2018). Although therapist expectations of treatment outcomes are not a specific change process or mechanism of change for either treatment, they can act as a common factor potentially influencing outcomes, akin to a placebo effect. Thus, if therapists in condition A expect their treatment to be more effective than therapists in condition B do, this may generate an advantage to treatment A (Falkenström et al., 2013). Given that ICBT was more established, it might have generated higher expectations for its effects. Conversely, the novelty and enthusiasm surrounding the newly developed IPDT could also lead to high expectations. Therefore, I adopted a non-directional hypothesis, positing a difference between the treatments in how expectations influenced outcomes.

6. Method

The sample comprised 272 adolescents aged 15–17 years ($M = 17.32$, $SD = 1.26$, 83% female) diagnosed with major depressive disorder, as established using the MINI International Neuropsychiatric Interview 7.0 (Sheehan et al., 1998). The patients were treated by 38 student therapists enrolled in their final year of a five-year psychologist MSc program. The study's primary outcome measure was the 17-item self-report version of the Quick Inventory of Depressive Symptomatology for Adolescents (QIDS; Bernstein et al., 2010), administered pre- and post-treatment, as well as weekly during treatment (totaling 12 repeated measurements). Additionally, I used a measure of therapist expectations, administered weekly during treatment (10 repeated measurements). This measure (Moras & Jones, 1992) consists of a single item (“How much improvement do you expect that your patient will experience as a result of treatment?”), with therapists responding on a 7-point Likert scale ranging from “-3” (“I expect my patient will feel much worse”) to “3” (“I expect my patient will feel much better”). Further details about the sample, therapists, and measures are available in the original report (Mechler et al., 2022).

7. Results

7.1. Dynamic panel bias

I began by estimating a two-level MLM model⁷ using the following equation:

$$QIDS_{it} = \mu + \alpha_Y(QIDS_{i,t-1} - \overline{QIDS_i}) + u_i + \varepsilon_{it} \quad (7)$$

This model defines $QIDS_{it}$ as a function of its lagged and person-mean centered version ($QIDS_{i,t-1} - \overline{QIDS_i}$). Estimated using the nlme package in R (Pinheiro et al., 2021), the autoregression coefficient $\hat{\alpha}_Y$ was found to be 0.52 ($p < .001$). Knowing that this value is subject to dynamic panel bias, and with Eq. (6) available, we can calculate the biased-corrected value. Following this equation, the relationship between the biased estimate $\hat{\alpha}_Y$ and the unbiased α_Y is expressed as $\hat{\alpha}_Y = \alpha_Y - \frac{(1+\alpha_Y)}{(T-1)}$. Substituting the values for $\hat{\alpha}_Y$ and T yields $0.52 = \alpha_Y - \frac{(1+\alpha_Y)}{11}$. After simplifying, we find that $\alpha_Y = 6.73/10 = 0.673$, suggesting an unbiased estimate for α_Y should be 0.673.

Next, I applied a dynamic panel model (DPM; Allison et al., 2017; Williams et al., 2018) to the same data, using a setup similar to the previous MLM. This model resembles Fig. 2b, except that it does not include an X variable. To align with the previous MLM model (Eq. (7)), I constrained the autoregressions, means, and residual variances to be equal over time. The only exceptions were the mean and variance for QIDS at time = 0, which were freely estimated, similar to their treatment in the MLM model. Estimated using maximum likelihood in lavaan (Rosseel, 2012), the model yielded $\hat{\alpha}_Y = 0.673$ ($p < .001$), matching the bias-corrected value from the MLM analysis. This indicates that structural equation modeling is not affected by dynamic panel bias.

In the next step, I added the lagged and centered Expect (expectations) variable as a predictor, first in the MLM model:

7.1.1. MLM 1

$$QIDS_{it} = \mu + \alpha_Y(QIDS_{i,t-1} - \overline{QIDS_i}) + \beta_{XY}(Expect_{i,t-1} - \overline{Expect_i}) + u_i + \varepsilon_{it} \quad (8)$$

Estimating this model using MLM, the results were $\hat{\alpha}_Y = 0.47$ ($p < .001$) and $\hat{\beta}_{XY} = -0.01$ ($p = .93$). To compare this with the DPM, I

⁷ Although therapist effects need to be accounted for in many psychotherapy applications, within-person coefficients are generally unaffected by therapist differences (Falkenström, Solomonov, & Rubel, 2020).

estimated a version of the model shown in Fig. 2b, with several constraints added for similarity to the MLM model: 1) Autoregressions in Expect (α_X) was set to zero, 2) cross-lagged effects from QIDS to Expect (β_{YX}) set to zero, 3) means and variances of QIDS equal from time 1–11, 4) covariances between error terms for $QIDS_{i,t}$ and $Expect_{i,t}$ (ψ_{XY}) set to zero. The results showed $\hat{\alpha}_Y = 0.66$ ($p < .001$) and $\hat{\beta}_{XY} = -0.14$ ($p = .07$), indicating differences in both autoregression and cross-lagged effect compared to the MLM model, though in neither model was the cross-lagged effect significant.

7.2. Residual- and observation-level models

The constrained version of the DPM exhibited poor fit to the data ($\chi^2(242) = 876.20$, $p < .001$, RMSEA = 0.10, CFI = 0.87, SRMR = 0.11) suggesting that its estimates may be unreliable. Upon relaxing the four constraints mentioned, the model fit improved significantly ($\chi^2(218) = 385.44$, $p < .001$, RMSEA = 0.05, CFI = 0.97, SRMR = 0.09).⁸ To compare a residual-level with an observation-level model, I estimated the random intercept cross-lagged panel model (RI-CLPM; Hamaker et al., 2015, shown in Fig. 2a), using the same constraints as in the DPM (equal regression coefficients and covariances over time, with freely estimated intercepts and residual variances).

Although it is not possible to make an MLM model identical to the RI-CLPM or DPM, I ran a second MLM model with two modifications. First, I added a categorical Week variable (dummy variables for Weeks 3–10) to emulate the free estimation of intercepts in the SEM models. Second, I included a non-lagged version of Expect (i.e., $Expect_{i,t} - \overline{Expect}_i$) to mimic the same-time residual covariances found in the SEM models:

7.2.1. MLM 2

$$QIDS_{i,t} = \mu + \alpha_Y(QIDS_{i,t-1} - \overline{QIDS}_i) + \beta_{XY}(Expect_{i,t-1} - \overline{Expect}_i) + \beta_{XY2}(Expect_{i,t} - \overline{Expect}_i) + Week_t + \varepsilon_{i,t} \quad (9)$$

The left-hand part of Table 1 displays model fit for the SEM models and information criteria for the MLM models. The information criteria are suitable for comparing within MLM or SEM models, but not between them, due to differences in data treatment and dependent variables used. Among the MLM models, MLM 2 demonstrated the best fit according to AIC and BIC. This model estimated the autoregression effect $\hat{\alpha}_Y$ at 0.25 ($p < .001$), while the lagged effect of Expect was statistically non-significant ($\hat{\beta}_{XY} = -0.09$, $p = .45$). In comparing the residual-level RI-CLPM with the observation-level DPM, the observation-level model showed better fit (AIC_{RI-CLPM} = 17,824.25, AIC_{DPM} = 17,666.91, BIC_{RI-CLPM} = 18,015.36, BIC_{DPM} = 17,872.44). The DPM's better fit may be due to its greater flexibility in accommodating various shapes of data, as discussed earlier.

Both SEM models yielded higher autoregression estimates than the MLM models and a significant lagged effect of Expect (RI-CLPM: $\hat{\alpha}_Y = 0.71$, $p < .001$, $\hat{\beta}_{XY} = -0.96$, $p < .001$; DPM: $\hat{\alpha}_Y = 0.50$, $p < .001$, $\hat{\beta}_{XY} = -0.52$, $p < .001$). These results illustrate how different modeling frameworks can lead to contrasting conclusions, with MLM suggesting

no cross-lagged effect of therapist expectations, while SEM indicates that higher expectations predict symptom improvement.

7.3. Long-run effect of therapist expectations on depressive symptoms

From the model results, we can calculate the long-run effect of a stable one-unit increase in therapist expectations. Applying the formula for the long-run effect, $\frac{\beta}{1-\alpha}$, and using results from the best-fitting DPM, we get $\frac{-0.52}{1-0.50} = -1.04$. Thus, the long-run effect in this case is twice the size of the cross-lagged effect. In practical terms, it suggests that if therapists consistently held one-point higher expectations throughout treatment, patients would be expected to score, on average, 1.04 points lower on the depression scale. Given that the baseline standard deviation of the QIDS-A17-R is 3.38, this 1.04-point reduction corresponds to a Cohen's d of 0.31, which is generally considered a small effect. However, in this study, the non-inferiority margin was set at 0.30, meaning a difference of 0.31 could be crucial in determining whether non-inferiority is achieved.

7.4. Testing whether different treatments work by different mechanisms

To illustrate the two distinct approaches to differential mechanisms, I began with the DPM model, adding the treatment group as a predictor. There are two methods for including the treatment effect in this model. One involves regressing all observed variables on the treatment variable, and the other involves regressing both the random effects (intercept factors) and the observed variables at the first occasion on treatment (K. A. Bollen, personal communication, 12th August 2022). In this case, both setups yielded highly similar model fit (AIC_{observed} = 17,662.57, BIC_{observed} = 17,875.31, AIC_{random} = 17,662.99, BIC_{random} = 17,875.74), leading me to choose the model in which all observed variables were

regressed on treatment.

Following the arguments for testing moderation discussed in the previous section, I added interaction terms between $Treatment_t$ and $Expect_{i,t}$. Since SEM is estimated using data in wide format, this resulted in a total of 10 interaction terms added to the model. However, I constrained the effects of these to be equal over time, effectively reducing to only one extra parameter. To simplify the model, I utilized orthogonalized interaction terms (Little, Bovaird, & Widaman, 2006), formed by residualizing the interaction terms on their constituent variables. The model, estimated using maximum likelihood with robust standard errors, fit well according to fit indices (scaled $\chi^2(458) = 655.60$, $p < .001$, RMSEA = 0.04, CFI = 0.96, SRMR = 0.07). However, the interaction effect between $Treatment_t$ and $Expect_{i,t}$ was not statistically significant (coefficient = -0.15 , $se = 0.17$, $z = -0.88$, $p = .38$), indicating no evidence that the effect of therapist expectations on patient symptoms differed between treatments.

In a second step, I estimated a mediation model using bootstrapping to get confidence intervals. The model fit the data reasonably well ($\chi^2(238) = 407.81$, $p < .001$, RMSEA = 0.05, CFI = 0.97, SRMR = 0.09). The indirect effect of treatment on depression via therapist expectations ($Treatment_t \rightarrow Expect_{i,t-1} \rightarrow QIDS_{i,t}$) was statistically significant (coefficient = 0.08, $se = 0.03$, $z = 2.51$, $p = .01$, 95% CI [0.02, 0.15]). This result indicates a mediated effect of treatment on depressive symptoms via therapist expectations, suggesting that higher expectations in the

⁸ The rules of thumb for model fit suggests the Chi-square test should be non-significant, RMSEA < 0.05, CFI ≥ 0.95, and SRMR ≤ 0.10. Thus, there is still room for improving the fit of this model. This can be done either by trying out several different between-person specifications, for instance by adding random slope factors to allow for person-specific trends (Bollen & Curran, 2004; Curran et al., 2014; Falkenström et al., 2023) or by adding moving average terms (Zyphur, Allison, et al., 2019). However, for the purpose of this demonstration, this model fit well enough.

Table 1
Model fit information and parameter estimates from multilevel and structural equation models testing the effect of therapist expectations on next-week depressive symptoms.

Model	χ^2 (df)	RMSEA	CFI	SRMR	AIC ^a	BIC ^a	α_Y^b, p	β_{XY}^c, p	α_X^d, p	β_{YX}^e, p
MLM1	–	–	–	–	8639.43	8666.66	0.43, <0.001	0.03, 0.77	–	–
MLM2	–	–	–	–	8451.65	8527.83	0.25, <0.001	–0.09, 0.45	–	–
RI-CLPM	550.78 (222)	0.07	0.93	0.12	17,824.25	18,015.36	0.71, <0.001	–0.96, <0.001	0.47, <0.001	–0.06, <0.001
DPM	385.44 (218)	0.05	0.97	0.09	17,666.91	17,872.44	0.50, <0.001	–0.52, <0.001	0.38, <0.001	–0.03, <0.001

^a The AIC and BIC calculated from MLM are not comparable to AIC/BIC calculated from the SEM models, since the log-likelihood is based on different variables in the respective models.
^b α_Y = autoregression for depressive symptoms.
^c = cross-lagged effect of therapist expectations on depressive symptoms.
^d = autoregression for therapist expectations.
^e = cross-lagged effect of depressive symptoms on therapist expectations.

ICBT group contributed to better outcome in this condition.⁹
Finally, to calculate the total indirect effect of treatment on the final outcome via therapist expectations, as outlined in the previous section, I began by assessing whether the direct effect of $Treatment_{it}$ on $QIDS_{i,t}$ could be set to zero. This would simplify the calculation of the total indirect effect $Treatment_{it} \rightarrow Expect_{i,t} \rightarrow QIDS_{i,t+1}$. The constrained model showed non-significantly worse fit ($\Delta\chi^2(1) = 0.43, p = .51$), suggesting it was acceptable for use. In this model, there were 1023 indirect paths from $Treatment_{it}$ to $QIDS_{i,t+1}$. The total effect, the sum of all indirect paths, was calculated as 0.27 (95% CI [0.06, 0.52]). This indicates that the mean QIDS score at treatment end for patients in the IPDT group was 0.27 points higher, reflecting worse depressive symptoms, than if therapist expectations had been as high as in the ICBT group. Though this represents a small effect (Cohen's $d = 0.08$), it demonstrates how differential mechanisms of change can be explored using a mediation model.

8. Limitations

While cross-lagged panel models offer stronger causal inference possibilities than cross-sectional studies, cross-lagged panel models are not a panacea for establishing causal effects. The limitations in drawing conclusions from these models include: 1) the potential for unobserved time-varying confounding, and 2) misspecification of time-lags. The first issue can, to some extent, be handled by measuring theoretically suspected confounders and adding them as covariates to the model, although there will always be a risk that important unknown confounders have been left out. Instrumental variable regression (e.g., Maydeu-Olivares et al., 2020) is an option, though finding suitable instruments can be challenging. In cases with specific data distributions and large samples, instrument-free adjustment methods are available (e.g., Falkenström et al., 2023), but these haven't been fully adapted for cross-lagged panel models yet.
When lags are too long, there's a risk of overlooking causal effects, and they may instead be subsumed by same-time covariances. Muthén and Asparouhov (2023) have explored methods to determine whether cross-lagged or contemporaneous effects fit the data best. However, contemporaneous effects are less convincing for causal inference since they don't satisfy the temporality condition. On the other hand, too short lags are generally less problematic, as it is possible to incorporate higher-order lags into the analysis.

⁹ Since the path from *Treatment* to *Expect* is a between-person effect, this is an exception from the argument made previously that therapist effects do not need to be included in these kinds of models (Falkenström et al., 2020). In this case, multilevel SEM might be used, with a random intercept term added for the *Expect* variable. This may affect the standard error for the *Treatment* → *Expect* coefficient but should not affect the *Expect* → *QIDS* path.

9. Practical recommendations for researchers

When studying processes and/or mechanisms of change in psychotherapy using time-lagged panel models, I recommend employing SEM with data in wide format, particularly when the number of repeated measures is small. The minimum number of repeated measures required is three for the RI-CLPM, and four or more for most other models that differentiate within- and between-person processes. Although there is no strict upper limit for the number of repeated measures, SEM estimation tends to become challenging as the number of variables increases relative to the sample size. This challenge is particularly pronounced in psychotherapy research, where there are often large proportions of missing data for later assessments. An absolute upper limit occurs when the number of estimated model parameters exceeds the sample size N . Constraining parameters to be equal over time reduces the number of parameters to be estimated and is likely a realistic assumption in many psychotherapy applications. Such constraints can facilitate estimation and increase statistical power (Zyphur, Allison, et al., 2019). For more details on sample size requirements in SEM, see Wolf et al. (2013).
MLM should be considered as a last resort for studying time-lagged prediction models when SEM is not feasible. When employing MLM, it is essential to acknowledge its limitation in terms of inability to account for autoregression while simultaneously separating within-person from between-person variances. Due to this, any findings should be interpreted with caution. An exception exists with DSEM, which, as mentioned, is suitable for use when the number of repeated measures is large. The minimum number of repeated measures for DSEM is not definitively known; however, Schultzberg and Muthén (2018) suggest that under certain conditions, as few as 10 may be feasible.

10. Future directions

There are several more advanced developments within SEM that have not been covered in this tutorial. Models specifying linear (Curran et al., 2014) or non-linear (Bauldry & Bollen, 2018) trends can be specified, to further minimize the risk of spurious cross-lagged effects due to confounding factors (Falkenström et al., 2023). Moving average terms can be added to accommodate more complex autoregressive structures (Asparouhov & Muthén, 2023; Zyphur, Allison, et al., 2019). Continuous time SEM is also an area of research that may improve psychotherapy mechanisms of change research, by analyzing effects as continuous influences rather than discrete effects between specified time-periods (Driver et al., 2017; Moggia et al., 2023). All these important developments offer opportunities of more refined analysis but will also require considerably stronger skills in statistics to set up models and interpret results.
In my experience, a particular weakness of wide-format SEM lies in its capacity to estimate interaction effects and/or random coefficients. Such models require computationally intensive, high-dimensional numerical integration (Zyphur et al., 2019), which is time-consuming and often necessitates very large samples for convergence. Although

Bayesian estimation may offer a more feasible approach for these types of models (Ozkok et al., 2021), they can still be challenging to code and may require lengthy estimation times. Another issue that frequently presents challenges in SEM is the presence of strongly unbalanced data. In psychotherapy research, it is common to encounter samples exhibiting exponentially declining participant numbers with an increasing number of sessions. Since interactions, random coefficients, and unbalanced data are issues that are generally much more manageable in MLM, future methodological development could focus on addressing the problem of dynamic panel bias within MLM. As mentioned, DSEM provides a solution to dynamic panel bias; however, DSEM does not seem feasible for use with short panels.

11. Conclusions

In psychotherapy research, a fundamental goal is to understand how treatments effect change, that is, identifying mechanisms of change. This understanding is crucial for our ability to refine, streamline, and enhance treatments to optimize outcome in the shortest possible time. To this end, we require methods that not only align with our conceptualizations of how psychotherapy operates but also enable the most rigorous possible tests of these mechanisms. I believe that time-lagged panel models meet these criteria. However, a recent comprehensive review of mechanisms of change research in adolescent psychotherapy revealed that almost two-thirds of the studies employed methods incapable of establishing temporal precedence (Taubner et al., 2023).

In this paper, I have discussed three often neglected issues in psychotherapy research related to cross-lagged panel models: 1) understanding dynamic panel bias, 2) calculating long-run effects, and 3) exploring differential mechanisms between treatments. One likely reason for this neglect is the predominant training in MLM rather than SEM among psychotherapy researchers. While SEM historically originated from Wright (1921) path analysis, which has strong connections to causal modeling (Bollen & Pearl, 2013), MLM is fundamentally a regression model enhanced with variance partitioning to handle clustered data (Raudenbush & Bryk, 2002). Given that latent variable modeling is the best known solution to dynamic panel bias, SEM emerges as the more suitable framework for studying time-lagged effects. Embracing SEM necessitates a shift in thinking towards causal models, characterized by interconnected equations that demonstrate the flow of presumed causal effects.

Long-run effects can be used to address the paradox that the same patients may show the strongest short-term effects of a mechanism, but still the poorest long-term outcome (Crits-Christoph & Connolly Gibbons, 2021). While short-term effects may lead to long-term consequences, this notion is based on a counterfactual interpretation of the estimated coefficients. In the applied research example, ICBT and IPDT showed similar factual outcome. However, a counterfactual interpretation suggests that IPDT patients would have fared slightly better if therapists in this condition had held more positive outcome expectations. Although this may not have significantly impacted the study's results, it exemplifies the interpretation of a mediational model.

Differential mechanisms between treatments have typically been studied using moderator models, which, as discussed, imply that treatments alter the causal relationships themselves. Mediation models, aligning more closely with typical conceptualizations of mechanisms of change, have not been extensively explored in psychotherapy research so far. Therefore, future psychotherapy research employing cross-lagged panel models should explore mediation models. I would also recommend explicitly stating the causal assumptions behind any model. For instance, if a moderation effect is being tested, the theoretical justification for believing that the moderator directly affects the causal relationship should be clearly stated. This approach could yield a more nuanced understanding of how various treatments affect change, paving the way for more targeted and effective therapeutic interventions.

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During the preparation of this work the authors used ChatGPT-4 for proofreading. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Data availability

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Appendix A. Supplementary data

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